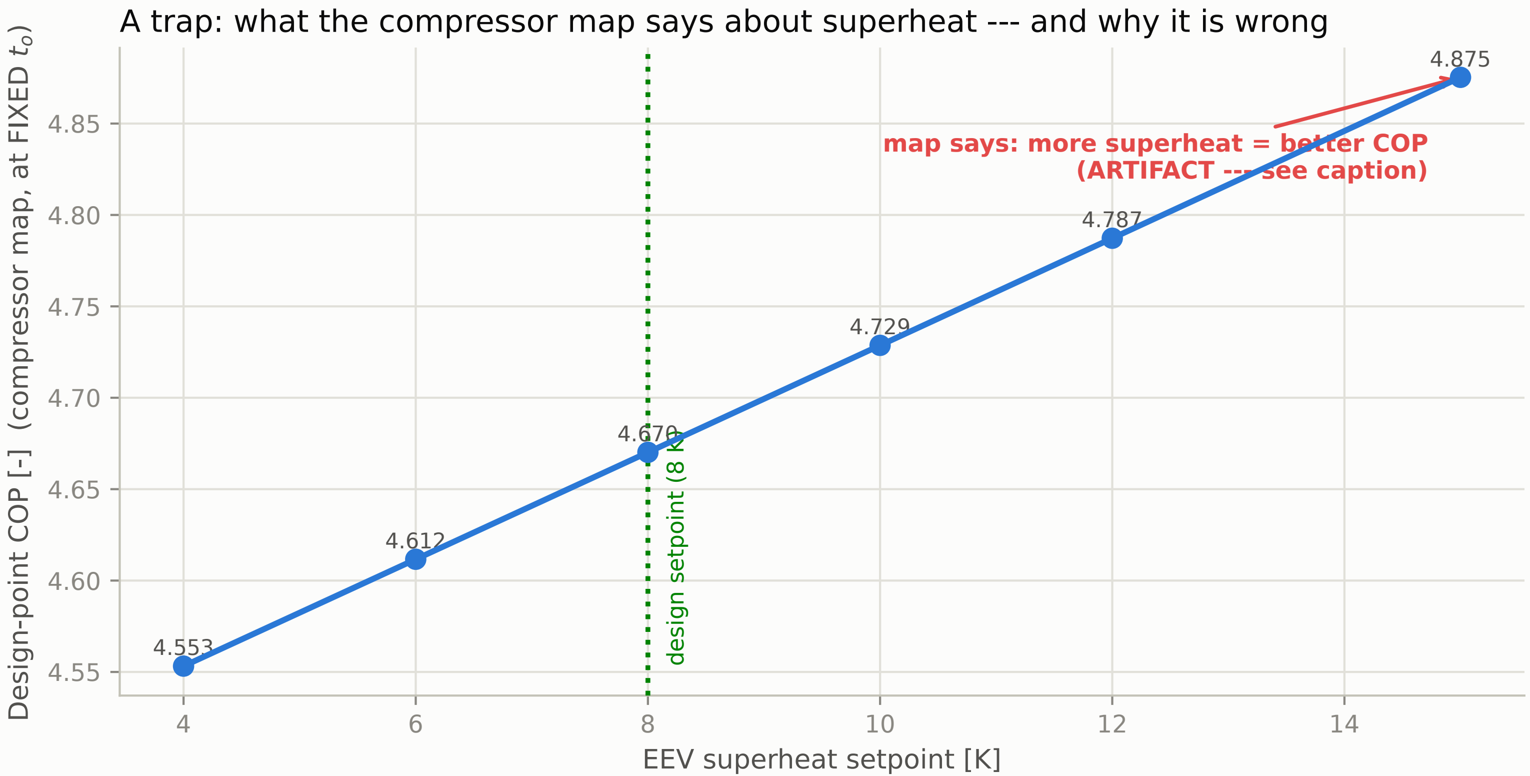

The map credits ALL superheat as USEFUL (EN12900 convention), so at FIXED t_o a higher setpoint raises h_1 and COP (6.8% across 4-15 K). But evaporator area is finite: the superheat tail has a poor gas-side coefficient and consumes area disproportionately, so demanding more superheat steals area from boiling and forces t_o DOWN --- costing more COP than the higher h_1 gains. The real trend is the opposite.